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Online detection of potato drying stages based on improved YOLOv7-tiny model

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ABSTRACT
To realize accurate online identification of different stages of the agricultural product drying process and overcome the limitations of empirical models, this study proposes a method for online identification of agricultural product drying stages based on machine vision, which enhances the YOLOv7-tiny model by adding an attention mechanism module to the feature layer and the up-adoption process. The recognition results were compared and evaluated with those of other versions of YOLO, Faster R-CNN, SSD, EfficientDet, and an unimproved YOLOv7-tiny network. The results showed that the average recognition accuracy of this method for the constant drying stage, first drying stage deceleration and second drying stage deceleration of potato slices reached 98.8%, which was superior to that of the model without the attentional mechanism module. This lays the foundation for the establishment of an on-line adaptive drying model for agricultural products.

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Drying; state detection; YOLOv7-tiny; potato; deep learning

1. Introduction

Currently, along with global warming and the increasing strain on natural resources, providing sufficient quality food for a continuously growing population has become a major challenge for mankind,[1] and a large amount of food and vegetables are lost due to rotting and deterioration due to the lack of loss-reducing technologies that are compatible with agricultural products in the process of transportation and storage. For example 30–40% of fruits and vegetables are wasted annually in some developing countries (e.g., South African region),[2] and nearly one-third of agricultural products are lost globally every year.[3]

Drying is an important and common agricultural product processing method,[4] that reduces microbial activity by lowering the water content of the material to prolong storage time. At present, research on the drying of agricultural products by scholars worldwide mainly focuses on energy consumption and product quality.[5–7] Drying accounts for 20–25% of food processing, and is also one of the most energy-consuming methods in industrial process.[8–9] Therefore, a 1% reduction in energy consumption may increase profits by 10%.[5,10,11] The drying principle is shown in Figure 1. During the drying, preheating, and constant-drying stages, the external diffusion rate v_1 is less than the internal diffusion rate v_2.[12] The drying rate of this process remains basically unchanged and is mainly controlled by the external environment. When $v_1 > v_2$, the process enters the deceleration drying stage (including the first drying stage deceleration and the second drying stage deceleration), and, a layer of dry area is formed on the surface of the material. This not only leads to hardening of the surface of the material, but also increases the resistance of the internal moisture migration to the outside.[13] The drying rate at this stage is mainly controlled by the internal diffusion rate of the material, however, it also involves energy consumption and quality issues. Savitha et al.[14] indicated that during the initial stage of moisture loss, tissues and cells undergo contraction, and as dehydration progresses, the pores eventually collapse, leading to the onset of a falling rate period. At this stage, the drying rate is mainly controlled by the internal moisture migration rate of the material,

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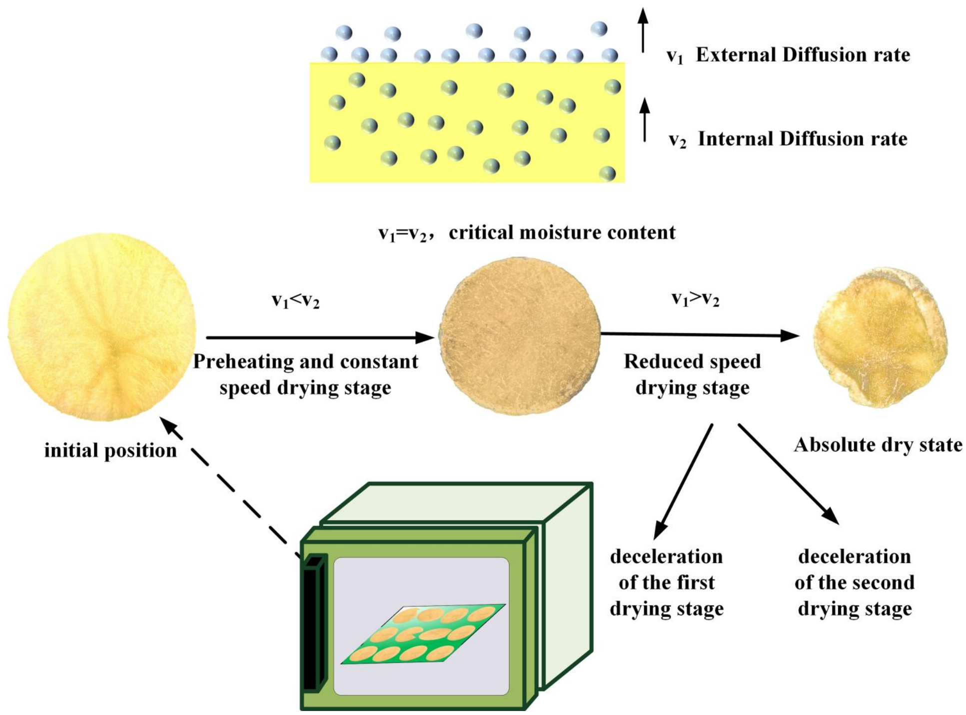

Figure 1. Schematic diagram of moisture movement during drying process.

which is the focus of research on energy consumption and quality.[13]

Potatoes are the fourth largest crop after corn, rice and wheat[15] and contain many micronutrients such as vitamins, minerals, phenolic compounds and high levels of dietary fiber.[16] In addition, potatoes are rich in resistant starch, which can promote blood glucose, prevent colon cancer, and inhibit fat accumulation,[17–18] They are usually dried and used as an ingredient in functional foods. Different strategies are adopted for the different drying stages; variable-condition drying is an important way to improve drying quality and reduce energy consumption.[2,19] Setiady et al.[20] studied microwave drying of potato tubers at 50, 60, and 70 °C. The results indicated that potatoes achieved a lower moisture content in a shorter time at 70 °C, but the color of dehydrated potatoes was severely negatively affected by overheating. In contrast, drying at 50 °C resulted in an acceptable color, though it required the longest drying time. Leeratanarak et al. [21] also confirmed that drying potatoes at lower temperatures achieved better color retention and reduced the browning index, although the drying time was extended. Bondaruk et al.[22] investigated the effects of drying conditions on the color, starch content, sugar content, mechanical properties, and microstructure of dried potatoes and found that the drying process caused the deformation and disintegration of cell walls and starch granules. Under hot air-drying conditions, the intensity of the structural changes depended on the drying temperature, with higher temperatures causing greater damage to the microstructure of potato cubes.

The premise of variable-condition drying is to accurately identify the drying state of the material, that is, the drying stage. Some researchers have studied the state of the material drying process, such as the water content. For example, Zhou et al.[23] successfully used hyperspectral technology to detect the moisture content of dried carrots. Cho et al.[24] used hyperspectral imaging technology to monitor the moisture content of the persimmon drying process and established a partial least squares regression prediction model with an R^2 of 0.9673 and a moisture measurement accuracy of 95%. This was limited by the long duration of HSI data acquisition, slow processing, high instrument cost, and difficulty in online detection.[25] Shukla et al.[26] used near-infrared (*NIR*) spectroscopy to measure the density, intensity, and moisture content of wood, and used partial least squares regression to establish a calibration model between the measured results and NIR data, which could be used to quickly and accurately estimate the characteristics examined in testing and evaluation procedures for commercial value. Machine vision, as a simple, reliable, nondestructive, and economical method, is considered as an online inspection technology that can be used during the drying process.[27] Recent studies have successfully applied deep learning- based target inspection techniques in various fields. For example, Wang et al.[28] demonstrated that machine vision can be used for the nondestructive inspection of moisture content in corn, and Wang et al.[29] used the improved Yolo-X model to identify and count the biological characteristics of hard clam larvae, with good results. Liu et al.[30] outlined the latest advances in deep learning-based object detection technology, demonstrating its superiority over traditional methods. However, few studies have been conducted on the use of machine- vision methods to determine the drying stage of agricultural products. The YOLO series algorithms are a type of algorithm that predicts target bounding boxes and the category of the target directly from the image to achieve a balance between detection accuracy and speed.[30] Among these, the YOLOv7 series developed by the Institute of Information Science, Academia Sinica, Taiwan surpasses the known object detectors in terms of recognition speed and accuracy.[31]

In this study, a detection algorithm based on an improved YOLOv7-tiny model is proposed. To improve its performance, a lightweight attention mechanism module was integrated into the feature layer and feature fusion output layer, extracted by the backbone of the YOLOv7-tiny model. The main purpose of this study is to establish an in-line and nondestructive detection model that can identify the drying state of potatoes. To evaluate the effectiveness and accuracy of the proposed model, a comparative test was conducted using a self-created dataset, and the test results were compared with those of other existing networks to provide a basis for determining the optimal recognition method.

2. Materials and methods

2.1. Image dataset

2.1.1. Experiments and samples

Potatoes were used as the experimental materials in this study. Approximately 40 fresh potatoes used in this experiment were purchased from local supermarkets in Tianjin. The potatoes were washed, peeled, and sliced, into disks with diameters of 50 and 38 mm, and a thickness of 5 mm ± 1 mm using a chopper and a round cutter. The slices were then dehydrated in a hot air-drying oven (Li-Chen Scientific Instruments, Hunan, China). Preliminary experiments revealed that under 70 °C hot air-drying conditions, it took about 190 min for the potato slices to dehydrate completely. In the formal experiment, the drying oven was preheated to 70 °C, and the mass and images of the potato slices before drying were collected. The slices were then quickly placed on the tray inside the drying oven, and at 15 min intervals, they were removed for the rapid collection of mass and image information until the drying was complete.

Each time the information was collected, each potato slice was individually measured. Four to five potato slices were used for each drying experiment, amounting to 120 slices in total, consisting of 60 slices with diameters of 38 mm and 60 slices with diameters of 50 mm, which resulted in 1441 sets of mass and image data.

2.1.2. Image acquisition

The image acquisition system built in this study is shown in Figure 2, which mainly consists of an LED light source, a dimmer, and a camera. The image acquisition process is completed in a closed black box to avoid interference form stray light. As described in Section 2.1.1, images of each potato slice were collected, including images before drying and at 15-minute intervals during the drying process until completion. After capturing the images, each potato slice was assigned a unique number and sorted according to drying time for subsequent grouping. A total of 1441 JPG format RGB images with a resolution of 2992 × 2992 pixels were collected.

2.1.3. Drying rate and moisture content

Under constant temperature drying conditions, the final dry state mass was taken as the adiabatic mass, the dry basis moisture content was calculated according to Equation (1), and the drying rate was calculated as shown in Equation (2).

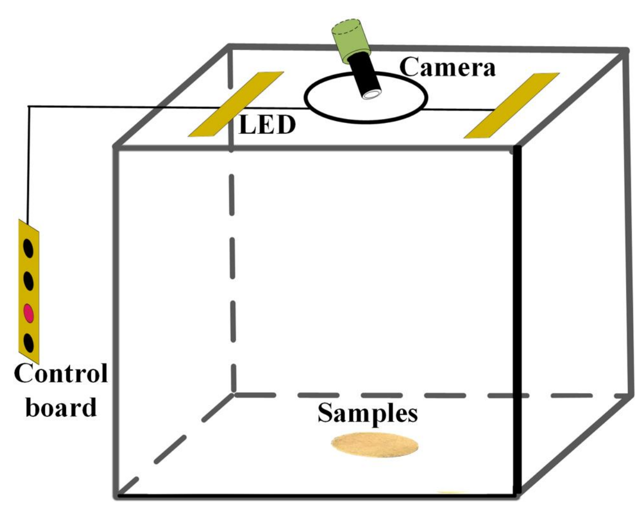

Figure 2. Image acquisition device.

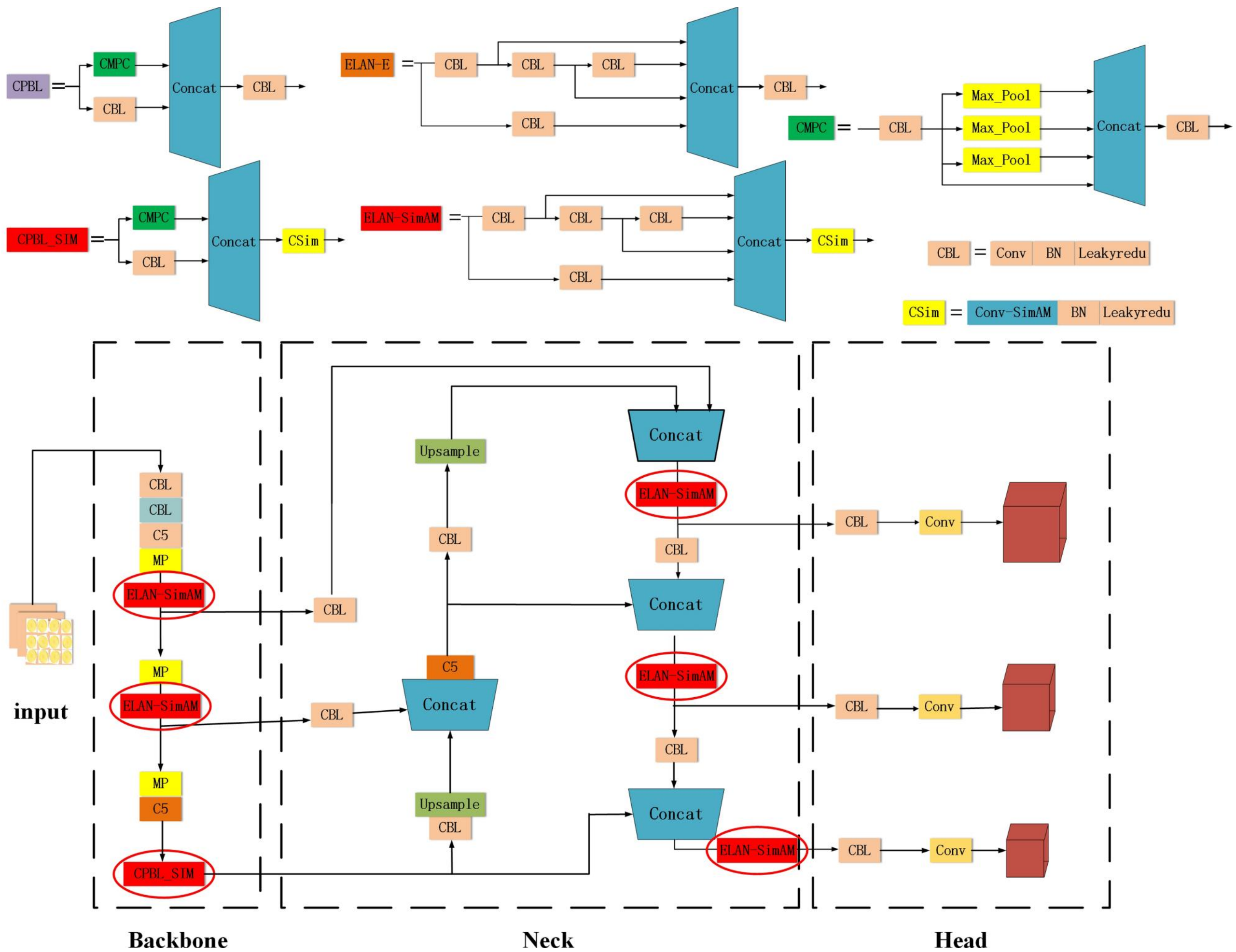

Figure 3. Structural diagram of the improved yolov7-tiny network.

$$M = \frac{m_i - m}{m} \times 100\% \tag{1}$$

Where M is the dry basis moisture content, %; m is the absolute quality, g; and m_i is the mass at moment i, g.

$$U = -\frac{\mathrm{dM}}{dt} \tag{2}$$

Where U is the drying rate, %/*min*; and t is time, *min*.

The relationship between the moisture content (M) and time (t) is called the drying curve, and that between the drying rate (U) and moisture content (M) is called the drying rate curve.

2.1.4. Computer configuration

This experiment was based on the *PyTorch* deep learning framework, the hardware platform for this experiment was *NVIDIA GeFore RTX 3060*, the operating system was *Windows 10 Professional*, and *Anaconda Python3.8* was used to train the algorithm on a personal computer using an *Intel(R) Core (TM) I7-11800 CPU @2.30 GHz* processor and *NVIDIA GeForce RTX 3060 Laptop GPU* to train the algorithms on a personal computer.

2.2. Model development

2.2.1. Improved YOLOv7-tiny model

YOLOv7-tiny consists of a backbone, neck, and head, and is characterized by its simplicity, fewer parameters than comparable models, and high computational efficiency. However, the feature extraction capability of the model is relatively weak. The network structure of the improved YOLOv7-tiny model is shown in Figure 3, where the improved components are marked with red ovals. Two ELAN-E modules and a CPBL module in the backbone were integrated with the lightweight attention mechanism SimAM to enhance the feature extraction capability of the model. In a addition, the ELAN-E model before the head incorporates the SimAM attention mechanism, which helps improve the information extraction ability for large, medium, and small objects.

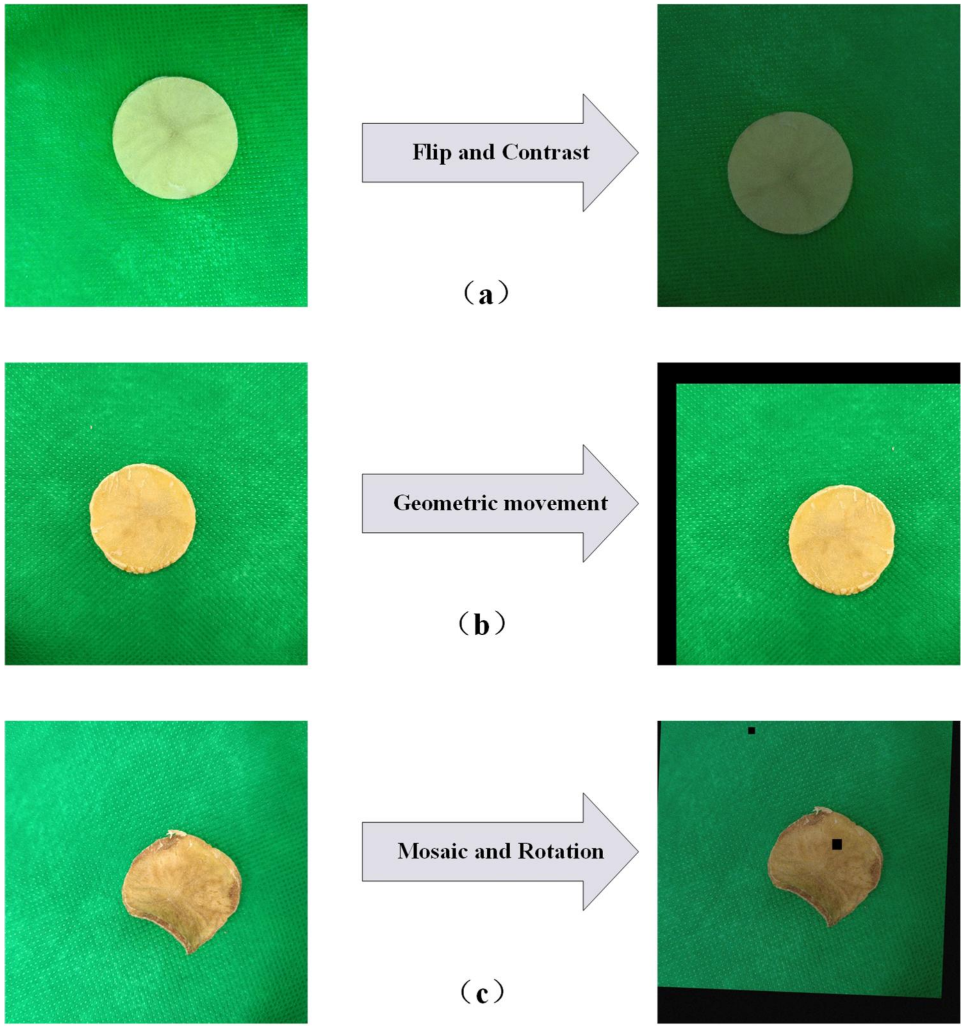

Figure 4. Data enhancement. (a) Flip and Contrast (b) Geometric movement (c) Mosaic and Rotation.

Table 1. Changes in mass before and after drying.

Diameter	50mm			38mm		
——	Max	Min	Avg	Max	Min	Avg
Initial Mass.	11.696g	9.676g	10.649g	7.092g	5.122g	5.902g
Final Mass.	2.445g	1.603g	1.997g	1.456g	1.078	1.078g

During the training process, the potato slice images were automatically resized to 640×640 pixels and then divided into an $S \times S$ grid. Each grid was responsible for predicting objects whose centers fell within that grid. Each grid predicted B bounding boxes along with their confidence scores, while outputting a probability distribution related to the object classes. Finally, the model calculated the loss based on these predictions and performed backpropagation to optimize the parameters. The network progressively extracts features from the images through several convolutional layers and activation functions, whereas the ELAN-SimAM dynamically adjusted the importance of different feature channels during feature extraction. This ensured that important features were emphasized, leading to a higher attention of the model on prominent features.

2.2.2. Data enhancement

To enhance the robustness of the model, data augmentation techniques such as mosaicking, contrast enhancement, and geometric transformations[32] were applied to the dataset, as shown in Figure 4. Each of the original 1441 captured images was augmented to generate two additional images, resulting in a final dataset of 4323 images. Subsequently, based on the experimental data processing, the images were classified into three categories: constant speed drying stage (CS), first falling rate drying stage (J1), and second falling rate drying stage (J2). The dataset was divided into training (train), validation (val), and testing (test) groups in an 8:1:1 ratio for the deep learning model training.

The K-means algorithm clustered the truth boxes of the dataset to obtain anchors suitable for the detecting the of potato drying stage.

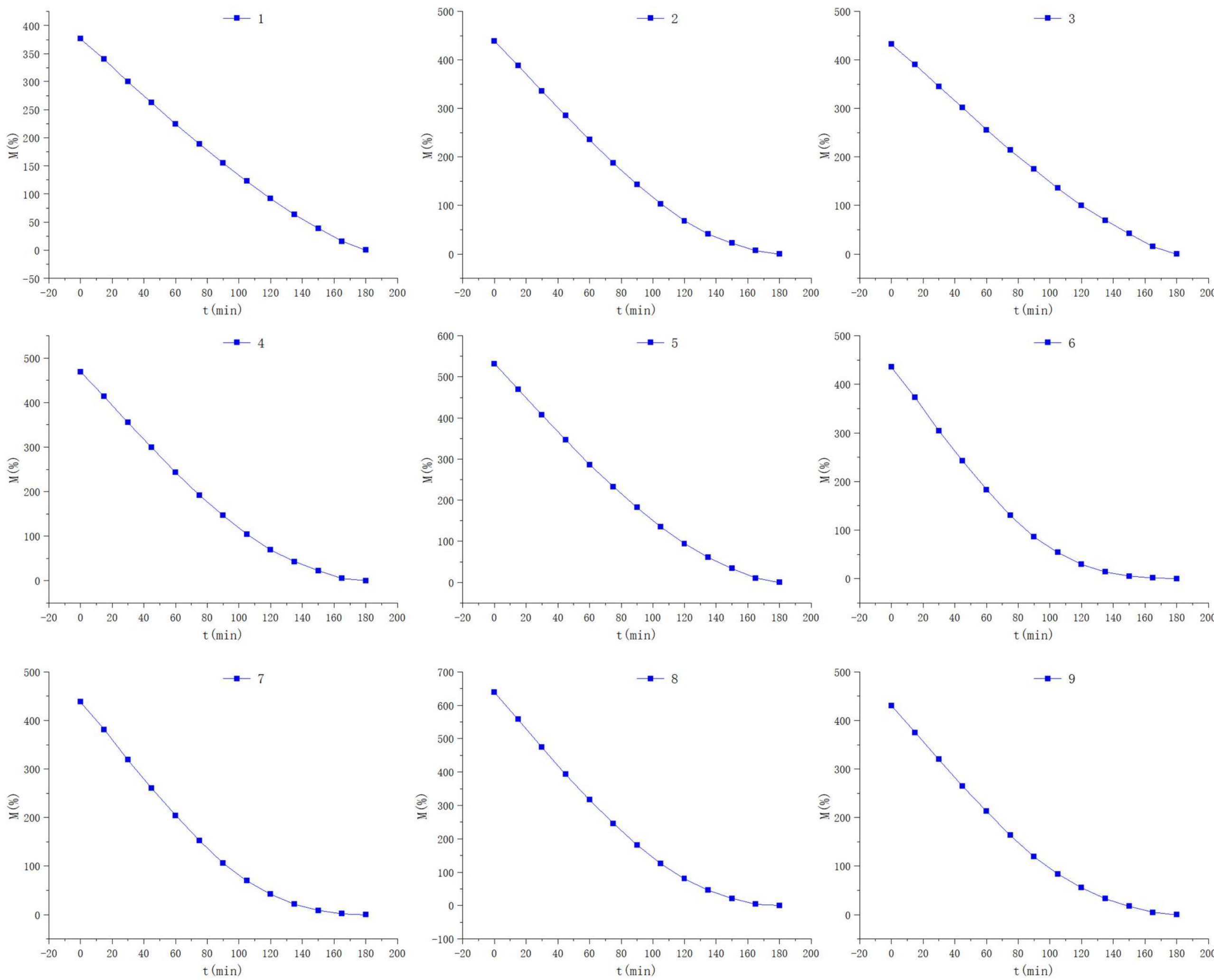

Figure 5. Partial results of drying curves.

2.2.3. SimAM attention mechanism

SimAM is a conceptually simple but very effective attention mechanism module, compared with the existing channel attention module and spatial attention module, SimAM can infer the three-dimensional attention weights of layer feature mappings without adding parameters to the original network.[33] The attention mechanism module further derives a fast closed solution to the capability function and shows that the solution can be implemented in just under ten lines of code. This lightweight module can significantly reduce the complexity and time required for model training. Another advantage is that most operators are selected based on the solution of the defined energy function, thereby avoiding excessive structural tuning.

2.2.4. Model evaluation indicators

Six evaluation metrics were used for complete coverage and unbiased analysis of the model performance: Precision (P), Recall (R), Average Precision (AP), F_1, FPS, and Mean Average Precision (mAP), as shown in Equations (3) through (8), respectively:

$$P = \frac{TP}{TP + FP} \times 100\% \tag{3}$$

$$R = \frac{TP}{TP + FN} \times 100\% \tag{4}$$

$$AP = \frac{TP + TN}{TP + TN + FP + FN} \times 100\% \tag{5}$$

$$F_1 = \frac{2PR}{P + R} \times 100\% \tag{6}$$

$$FPS = \frac{1}{Nms + Inf} \tag{7}$$

$$mAP = \frac{\sum_{i=1}^{N} AP_i}{N} \tag{8}$$

In Equations (3)–(5), *TP* denotes a true case, that is, a positive case that is correctly categorized; *TN* denotes a true negative case, that is, a negative case that is correctly categorized; *FP* denotes a false

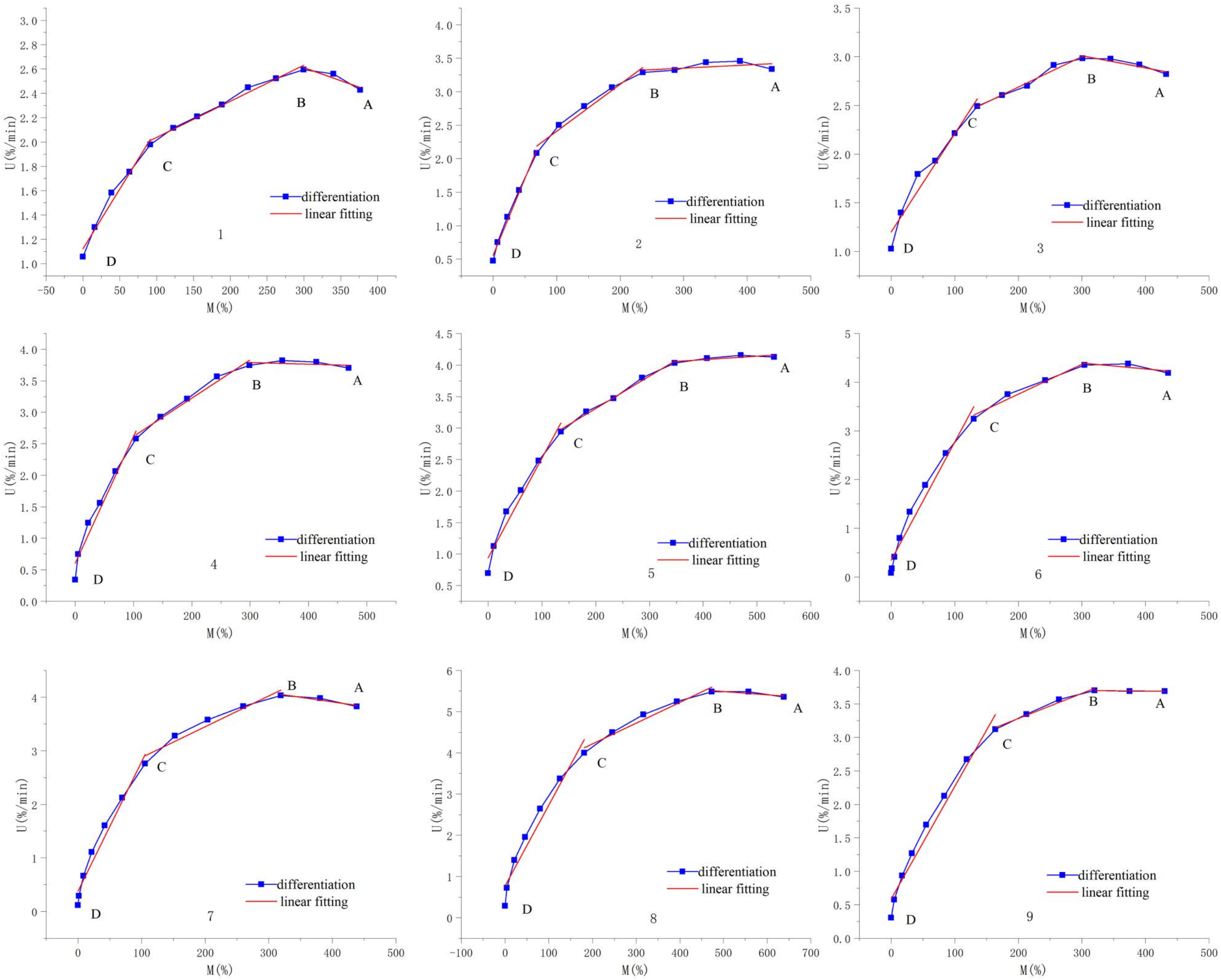

Figure 6. Partial results of drying rate curves.

positive case, that is., an originally negative case that is incorrectly categorized as a positive case; *FN* (denotes a false negative case, that is, an originally positive case that is incorrectly categorized as a negative case. In Equation (7) Inf is the speed of inference; and Nms is the post-processing time.

3. Results and discussion

3.1. Drying stage recognition

The changes in the mass of the potato slices before and after drying are shown in Table 1. Li et al.[34] concluded that smoothing the drying curve significantly reduced the fluctuations in the resulting drying rate curve when processing the drying quality information.

Information on the potato slice drying process was processed by calculating the moisture content at each time point according to Equation (1), and the M-t drying curve was plotted using Origin 2019 and smoothed using the Smooth function. The drying curves of nine randomly selected samples from the 120 potato slices are shown in Figure 5. Then, based on the smoothed drying curves and Equation (2), the drying rate curves were obtained. A curve fitting function was used to perform piecewise fitting of the drying- rate curves, as shown in Figure 6. The blue curve represents the drying rate curve obtained from differential separation, whereas the red line represents was the piecewise fitted drying rate curve. The data processing in this experiment was conducted on each potato slice individually rather than in batches; therefore, there are no error bars in the figure.

As shown in Figure 6, the drying rate curve can be divided into three stages: constant- rate drying, deceleration of the first drying stage (J1) and deceleration of the second drying stage (J2).

The images were labeled using the online annotation tool Make Sense and the samples were divided into three categories. In drying, the images

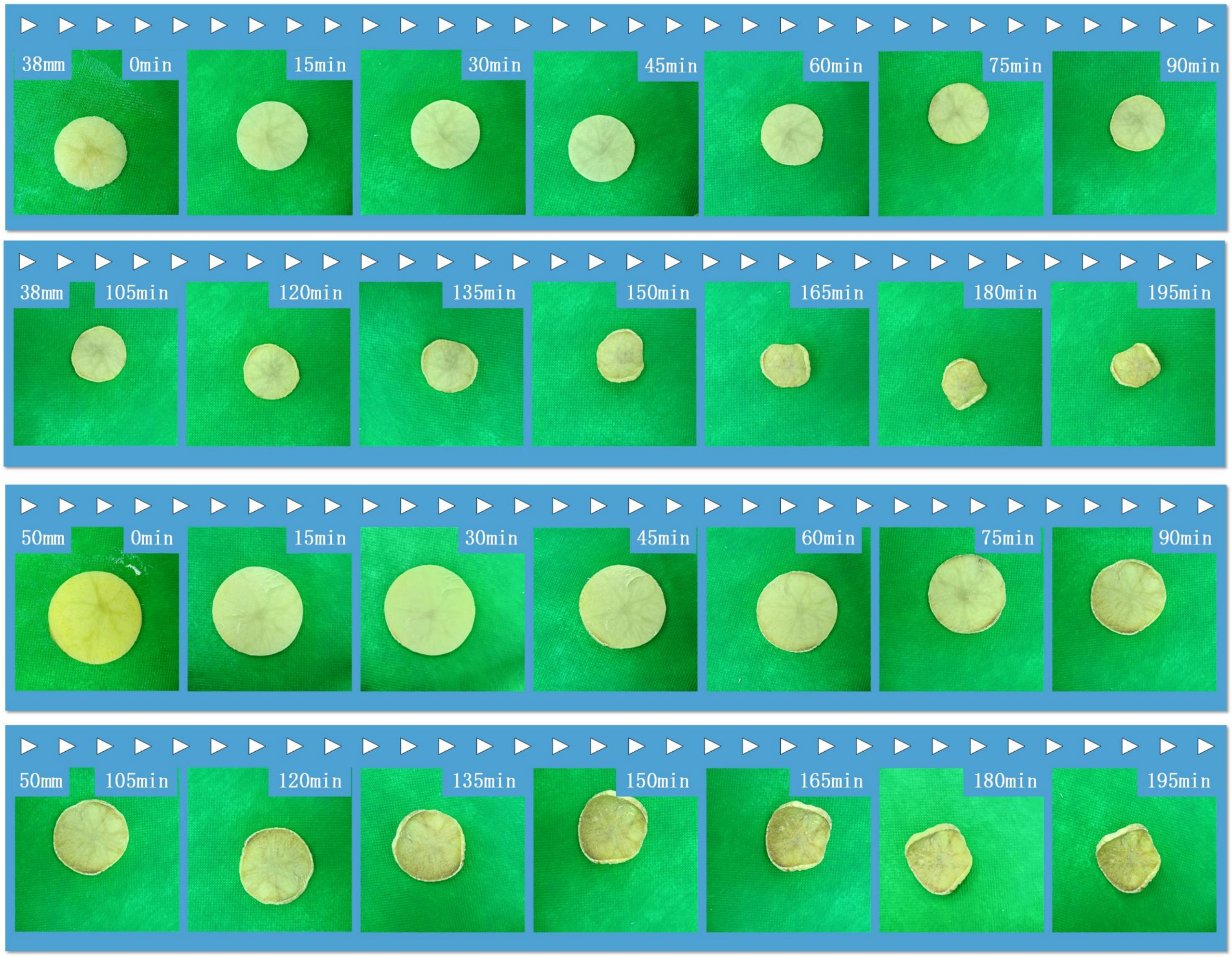

Figure 7. Image of potato slices during the drying process.

corresponding to the AB segment (constant- rate drying) were labeled as "CS," the images corresponding to the BC segment (deceleration of the first drying stage) were labeled as "J1," and the image corresponding to the CD section (deceleration of the second drying stage) is labeled as "J2." Each annotated image generated a dataset in .txt format that contains the drying stage classification of potato chips and the coordinates within the corresponding image.

3.2. Image analysis

Figure 7 shows the image changes of potato slices with two different diameters (50 and 38 mm) at different drying times. Combining Figures 5 and 7, the external changes in the two potato slice diameters under hot air -drying exhibit similar trends. As the drying time increased, the moisture content decreased, the external dimensions of the potato slices decreased, and curling occurred, along with color changes. This is attributed to the evaporation of moisture during the drying process and changes in the microstructure of the surface and interior of the potato slices.

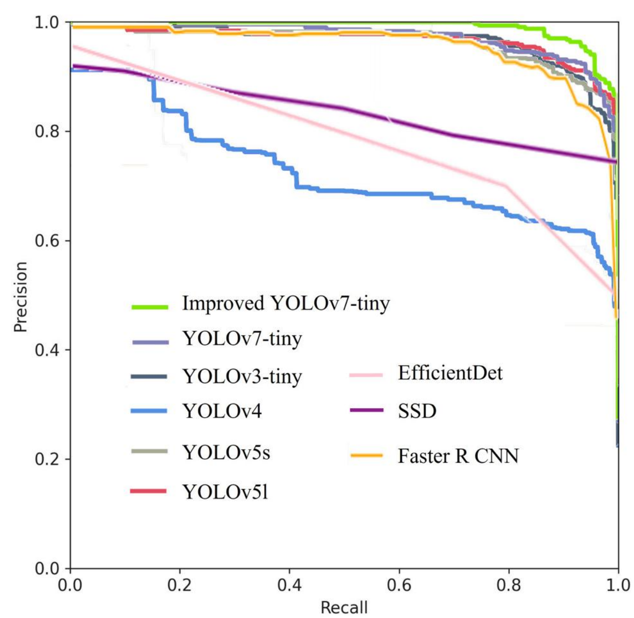

Figure 8. Comparison of P–R curves with different models.

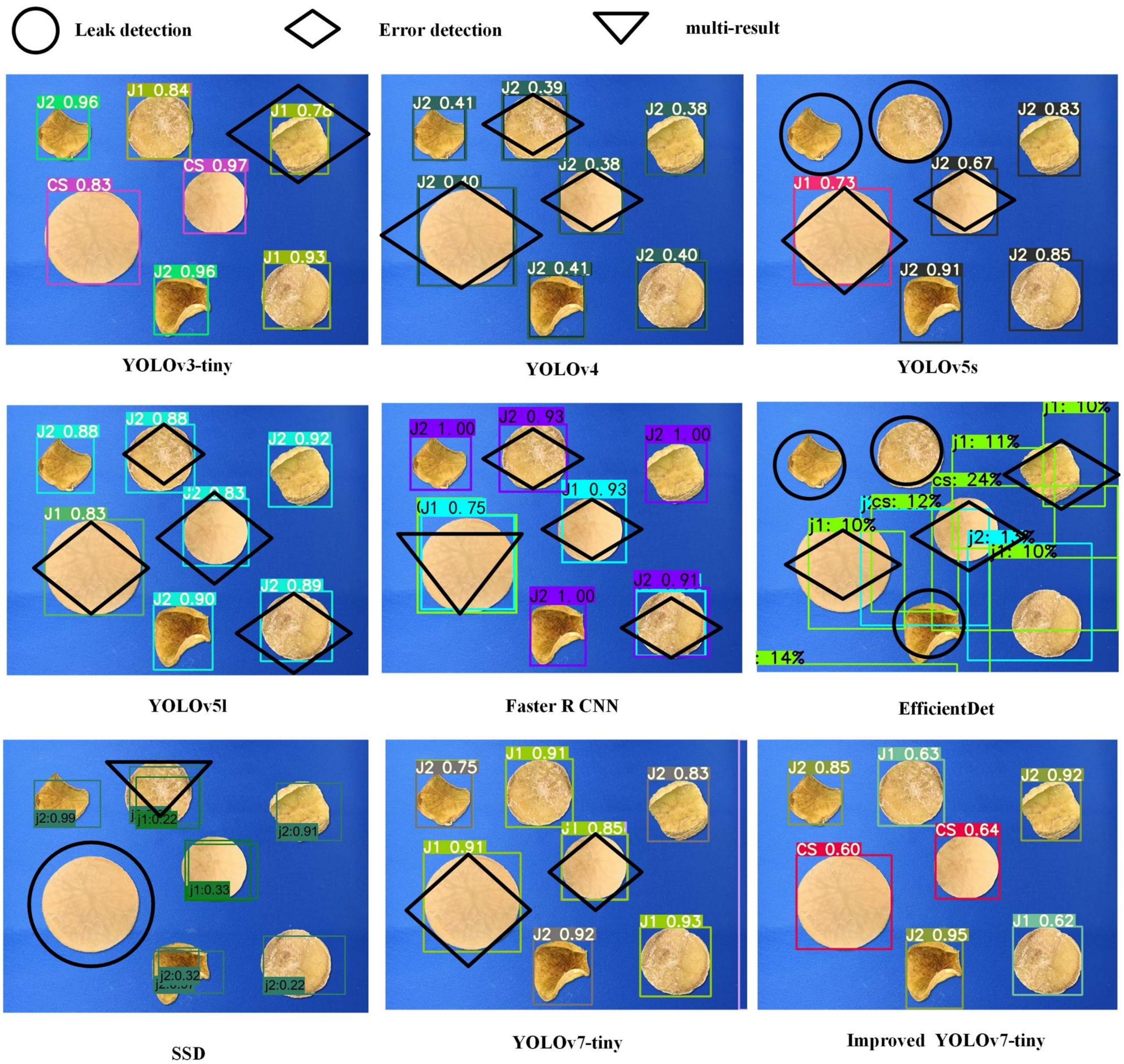

Figure 9. Comparison of experimental detection results for different models.

Specifically, in the initial stage of moisture loss, the tissues and cells underwent contraction, and subsequent dehydration led to cell collapse, resulting in pore collapse and, ultimately, the onset of the falling rate period.[14] As the drying time increased, higher temperatures caused greater damage to the microstructure of the potato cubes.[22]

3.3. Comparison with classical algorithm

This study compared the proposed model with other popular object detection models, including earlier versions of YOLO, SSD, Faster R-CNN, EfficientDet, and the unimproved YOLOv7-tiny model, for a comprehensive evaluation. Each network was trained for 180 iterations to ensure maximum convergence. The PR (Precision–Recall) curves of these models were shown in Figure 8, and it is evident that the PR curve of the improved YOLOv7-tiny model enclosed the largest area, indicating that it had the highest mAP.

Figure 9 shows the detection results of the different models for the same photo. In this study, potato slices with varying drying time intervals were detected in the same image with missed, false, or multiple detections marked. From Figure 9, it is clear that Faster R-CNN exhibited numerous error detections, whereas the SSD model showed low confidence and instances of missed and multiple detections. Other lower versions of YOLO also demonstrated varying degrees of errors and low confidence. In contrast, the improved YOLOv7-tiny model exhibited promising detection performance.

Table 2. Comparison of the accuracy results for the deep learning models.

	Recall(%)						
Method	CS (%)	J1 (%)	J2 (%)	F_1(%)	FPS	mAP (%)	Size (MB)
YOLOv3-tiny	99	66	82	91	208.3	96.4	16.6 MB
YOLOv4	37	28	44	59	62.1	72.7	100 MB
YOLOv5s	98	87	71	88	117.6	96.3	13.7 MB
YOLOv5l	99	82	82	91	95.2	96.6	89.4 MB
SSD	91	87	73	81	62	91.7	183 MB
EfficientDet	80	82	80	74.6	33.35	69.8	15 MB
Faster R CNN	83	65	85	90	10.83	96.2	108 MB
YOLOv7-tiny	99	85	74	90	222.2	97.5	11.7 MB
Improved YOLOv7-tiny	**99**	**89**	**83**	**95**	**243.9**	**98.8**	**11.7 MB**

Table 2 lists the detection performance parameters of nine models. Compared to the other models, the improved YOLOv7-tiny showed the best performance in terms of recognition speed (FPS) and outperformed the others in detection performance metrics such as F_1 score, recall, and mAP. This indicates that the improved YOLOv7-tiny model achieves a good balance between recognition speed and accuracy with significantly enhanced performance. In addition, the model size was only 11.7MB, which is comparable to that of the unimproved YOLOv7-tiny and smaller than those of the other models. This is related to the lightweight nature of SimAM, which positively affects the deployment of the model.

In summary, the improved YOLOv7-tiny model detection algorithm was advantageous for accurately identifying the drying stage of potato chips. It outperformed the other models by maintaining high accuracy and reducing missed detections and misidentifications.

4. Conclusion

In this study, online recognition of the constant speed drying stage, deceleration of the first drying stage and deceleration of the second drying stage of the hot air-drying process of potato chips was performed based on the machine vision method, and a three-dimensional attention mechanism module was added to further improve the accuracy of model recognition. The result showed that the improved YOLOv7-tiny model outperformed the networks including other versions of YOLO, SSD, Faster R CNN and EfficientDet, with the average mAP reaching 98.8%, which was higher than that of the unimproved model (97.5%), and a running speed of 243.9 FPS, which was 9.8% higher. The size of the model did not increase because of the increase in networking, which made it easier to carry on hardware devices. This provides an effective method for online identification of the material drying state during the drying process of agricultural products.

Disclosure statement

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References

[1] Aschemann-Witzel, J. Waste Not, Want Not, Emit Less. *Science* **2016**, *352*, 408–409. DOI: 10.1126/science.aaf2978.
[2] Zeng, S.; Du, Z.; Lv, W.; Li, D.; Su, D.; Lv, H. Experimental Study on the Hygrothermal Dynamics of Peanut (*Arachis hypogaea* Linn.) in the Process of Superposition and Variable Temperature Drying. *Drying Technol.* **2022**, *40*, 1463–1477. DOI: 10.1080/07373937.2021.1873359.
[3] Xue, L.; Liu, G.; Parfitt, J.; Liu, X.; Van Herpen, E.; Stenmarck, Å.; O'Connor, C.; Östergren, K.; Cheng, S. Missing Food, Missing Data? A Critical Review of Global Food Losses and Food Waste Data. *Environ. Sci. Technol.* **2017**, *51*, 6618–6633. DOI: 10.1021/acs.est.7b00401.
[4] Mujumdar, A. S.; Xiao, H. W. *Advanced Drying Technologies for Foods*. CRC Press: New York. **2019**.
[5] Delele, M. A.; Weigler, F.; Mellmann, J. Advances in the Application of a Rotary Dryer for Drying of Agricultural Products: A Review. *Drying Technol.* **2015**, *33*, 541–558. DOI: 10.1080/07373937.2014.958498.
[6] Lamidi, R. O.; Jiang, L.; Pathare, P. B.; Wang, Y. D.; Roskilly, A. P. Recent Advances in Sustainable Drying of Agricultural Produce: A Review. *Appl. Energy* **2019**, *233-234*, 367–385. DOI: 10.1016/j.apenergy.2018.10.044.
[7] Ertekin, C.; Firat, M. Z. A Comprehensive Review of Thin-Layer Drying Models Used in Agricultural

Products. *Crit. Rev. Food Sci. Nutr.* **2017**, *57*, 701–717. DOI: 10.1080/10408398.2014.910493.

[8] Kudra, T.; Martynenko, A. Energy Aspects in Electrohydrodynamic Drying. *Drying Technol.* **2015**, *33*, 1534–1540. DOI: 10.1080/07373937.2015.1009540.

[9] Strumillo, C. Perspectives on Developments in Drying. *Drying Technol.* **2006**, *24*, 1059–1068. DOI: 10.1080/07373930600778056.

[10] Zhou, X.; Wang, S. Recent Developments in Radio Frequency Drying of Food and Agricultural Products: A Review. *Drying Technol.* **2019**, *37*, 271–286. DOI: 10.1080/07373937.2018.1452255.

[11] Osae, R.; Essilfie, G.; Alolga, R. N.; Akaba, S.; Song, X.; Owusu-Ansah, P.; Zhou, C. Application of Non-Thermal Pretreatment Techniques on Agricultural Products Prior to Drying: A Review. *J. Sci. Food Agric.* **2020**, *100*, 2585–2599. DOI: 10.1002/jsfa.10284.

[12] Pai, R. K.; Sindhuja, V.; Ramachandran, P. A.; Thorat, B. N. Mass Transfer "Regime" Approach to Drying. *Ind. Eng. Chem. Res.* **2021**, *60*, 9613–9623. DOI: 10.1021/acs.iecr.1c01680.

[13] Yao, B.; Zhang, X. K.; Wen, X. D. Developments of Variable Temperature Drying Technology for Agricultural Products at Home and Abroad. *Food Sci. Technol.* **2015**, *40*, 94–98. DOI: 10.13684/j.cnki.spkj.2015.07.022.

[14] Savitha, S.; Chakraborty, S.; Thorat, B. N. Microstructural Changes in Blanched, Dehydrated, and Rehydrated Onion. *Drying Technol.* **2022**, *40*, 2550–2567. DOI: 10.1080/07373937.2022.2078347.

[15] Ahmed, S.; Zhou, X.; Pang, Y.; Xu, Y.; Tong, C.; Bao, J. Genetic Diversity of Potato Genotypes Estimated by Starch Physicochemical Properties and Microsatellite Markers. *Food Chem.* **2018**, *257*, 368–375. DOI: 10.1016/j.foodchem.2018.03.029.

[16] Bao, H.; Zhou, J.; Yu, J.; Wang, S. Effect of Drying Methods on Properties of Potato Flour and Noodles Made with Potato Flour. *Foods* **2021**, *10*, 1115. DOI: 10.3390/foods10051115.

[17] Bach, S.; Yada, R. Y.; Bizimungu, B.; Fan, M.; Sullivan, J. A. Genotype by Environment Interaction Effects on Starch Content and Digestibility in Potato (*Solanum tuberosum* L.). *J. Agric Food Chem.* **2013**, *61*, 3941–3948. DOI: 10.1021/jf3030216.

[18] Larder, C. E.; Abergel, M.; Kubow, S.; Donnelly, D. J. Freeze-Drying Affects the Starch Digestibility of Cooked Potato Tubers. *Food Res. Int.* **2018**, *103*, 208–214. DOI: 10.1016/j.foodres.2017.10.034.

[19] Yan, C.; Yubai, C. Experimental Study on Litchi Microwave Drying Characteristics and Its Effects on the Quality of Litchi. *Trans. Chin. Soc. Agric. Eng.* **2004**, *20*, 192–194. DOI: 10.3321/j.issn:1002-6819.2004.04.045.

[20] Setiady, D.; Clary, C.; Younce, F.; Rasco, B. A. Optimizing Drying Conditions for Microwave-Vacuum (MIVAC®) Drying of Russet Potatoes (*Solanum tuberosum*). *Drying Technol.* **2007**, *25*, 1483–1489. DOI: 10.1080/07373930701537187.

[21] Leeratanarak, N.; Devahastin, S.; Chiewchan, N. Drying Kinetics and Quality of Potato Chips Undergoing Different Drying Techniques. *J. Food Eng.* **2006**, *77*, 635–643. DOI: 10.1016/j.jfoodeng.2005.07.022.

[22] Bondaruk, J.; Markowski, M.; Błaszczak, W. Effect of Drying Conditions on the Quality of Vacuum-Microwave Dried Potato Cubes. *J. Food Eng.* **2007**, *81*, 306–312. DOI: 10.1016/j.jfoodeng.2006.10.028.

[23] Zhou, H.; Huang, M.; Zhu, Q.; Zhang, M. Developing C-LSTM Model for Evaluating Moisture Content of Carrot Slices During Drying. *Drying Technol.* **2022**, *40*, 2964–2974. DOI: 10.1080/07373937. 2021.1983822.

[24] Cho, J.; Choi, J.; Moon, K. Hyperspectral Imaging Technology for Monitoring of Moisture Contents of Dried Persimmons During Drying Process. *Food Sci. Biotechnol.* **2020**, *29*, 1407–1412. DOI: 10.1007/s10068-020-00791-x.

[25] Cheng, S.; Su, W.; Yuan, L.; Tan, M. Recent Developments of Drying Techniques for Aquatic Products: With Emphasis on Drying Process Monitoring with Innovative Methods. *Drying Technol.* **2021**, *39*, 1577–1594. DOI: 10.1080/07373937.2021.1895205.

[26] Shukla, S. R.; Sharma, S. K. Estimation of Density, Moisture Content and Strength Properties of Tectona Grandis Wood Using Near Infrared Spectroscopy. *Maderas-Ciencia Tecnol.* **2021**, 23, 1–12. DOI: 10.4067/s0718-221x2021000100418.

[27] Behroozi Khazaei, N.; Tavakoli, T.; Ghassemian, H.; Khoshtaghaza, M. H.; Banakar, A. Applied Machine Vision and Artificial Neural Network for Modeling and Controlling of the Grape Drying Process. *Comput. Electron. Agric.* **2013**, *98*, 205–213. DOI: 10.1016/j.compag.2013.08.010.

[28] Wang, R.; Han, F.; Jin, Y.; Wu, W. Correlation between Moisture Content and Machine Vision Image Characteristics of Corn Kernels. *Int. J. Food Prop.* **2020**, *23*, 319–328. DOI: 10.1080/10942912.2020.1720715.

[29] Wang, Y.; Wu, H.; Hua, X.; Ren, D.; Li, Y.; Mu, Z.; Xu, W.; Wei, Y.; Zhang, T.; Jiang, Y. Biological Characters Identification for Hard Clam Larva Based on the Improved YOLOX-s. *Comput. Electron. Agric.* **2023**, *212*, 108103. DOI: 10.1016/j.compag.2023.108103.

[30] Liu, H.; Ma, X.; Yu, Y.; Wang, L.; Hao, L. Application of Deep Learning-Based Object Detection Techniques in Fish Aquaculture: A Review. *JMSE* **2023**, *11*, 867. DOI: 10.3390/jmse11040867.

[31] Wang, C.; Bochkovskiy, A.; Liao, H. YOLOv7:Trainable Bag-of-Freebies Sets New State-of-the-Art for Real-Time Object Detectors. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. **2022**, pp 7464–7475. DOI: 10.48550/arXiv.2207.02696.

[32] Yue, X.; Wang, Q.; He, L.; Li, Y.; Tang, D. Research on Tiny Target Detection Technology of Fabric Defects Based on Improved YOLO. *Appl. Sci.-Basel* **2022**, *12*, 6823. DOI: 10.3390/app12136823.

[33] Yang, L.; Zhang, R. Y.; Li, L.; Xie, X. SimAM: A Simple, Parameter-Free Attention Module for Convolutional Neural Networks. Presented at International conference on machine learning, 2021, pp 11863–11874.

[34] Li, Z.; Liu, W. F.; Wang, Y. Discussion on Processing Methods of Drying Curve in Drying Experiment. *Chin. J. Chem. Educ.* **2022**, *43*, 49–56. DOI: 10.13884/j.1003-3807hxjy.2021060197.